

Name: AegisMesh: Cooperative Kinetic Safety Architecture

Problem Addressed: Open-cast mine haulage becomes highly dangerous or grinds to a halt during dense fog because drivers out-drive their visual stopping distance. Existing solutions fail by relying solely on easily blinded optical sensors, creating centralized network single-points-of-failure, or employing static distance alarms that ignore the physics of heavily loaded trucks on steep gradients.

Core Concept: AegisMesh is an offline-first, radar-primary safety mesh. It fuses vehicle-specific kinetic braking models with decentralized spatial awareness, allowing a fleet to collaboratively prove a "Safe-to-Proceed" corridor.

How it works: 77GHz radar acts as the primary distance measure, functioning independently of network availability. When a vehicle detects a hazard, an onboard AI packages the detection into a micro-payload Spatial Vector Packet and broadcasts it directly to peers. Approaching vehicles receive this vector, project it onto their own route graph, and use a Kinetic Priority Engine to negotiate right-of-way based on mass and gradient. If the required stopping distance exceeds the verified clear road ahead, the system issues a deterministic speed restriction.

2. System Architecture

Complete Architecture:

The architecture operates across four decentralized tiers to eliminate single points of failure.

- **Tier 1: Vehicle Safety Edge (Offline-First):** The primary safety loop resides entirely on the vehicle. It executes the radar sensing, state machine, and stopping margin calculations locally.
- **Tier 2: Swarm Peer-to-Peer (V2V):** Vehicles directly exchange Spatial Vector Packets, mass/gradient data, and routing intents over a localized RF mesh.
- **Tier 3: Adaptive Road Guardians:** Fixed nodes monitor fog trends at critical junctions. Idle trucks temporarily adopt this node role, utilizing their onboard sensors to expand the network's predictive coverage at zero additional hardware cost.
- **Tier 4: Passive Infrastructure:** Power-free patterned corner reflectors (RadarGlyphs) are placed at blind curves and bench edges to provide absolute localization in deep pits where GNSS signals degrade.

End-to-End Workflow:

1. **Sense:** Radar maps the corridor; dual-spectrum roadside sensors differentiate true fog from routine mine dust.
2. **Verify & Trust:** A multi-sensor Trust Engine aggregates hazard confidence, rejecting stale or unauthenticated peer packets.
3. **Calculate:** The BrakeTwin engine calculates the exact stopping distance needed based on the truck's current payload, road gradient, and validated surface friction.
4. **Coordinate:** At single-lane blind curves, vehicles request a Virtual Block Movement Token; the system fails-closed to prevent head-on collisions if communications drop.
5. **Actuate:** The driver receives a strictly advisory, singular recommended safe speed (e.g., \$15km/h\$) rather than an opaque "92% confidence" score.

3. Technical Design

- **Hardware:**
 - *Vehicles:* 77GHz FMCW Radar (primary), RGB camera (secondary/classification), RTK-GNSS + IMU, Edge AI compute (e.g., Raspberry Pi 5 / Jetson), and a safety MCU (ESP32/STM32) for watchdog and HMI rendering.
 - *Roadside Nodes:* ESP32, multi-modal discriminator sensors, LoRa transceiver.
 - *Worker Tags:* Low-power active RF beacons for pedestrian detection.
- **Sensors & Actuators:**

- **Fog vs. Dust Discriminator:** Uses a 940nm IR LED combined with a visible red LED. Fog heavily absorbs IR but passes red; dust reflects both. This prevents false alarms triggered by haul-road dust.
- **Actuation:** Strictly advisory LED/audio/haptic dashboard for the prototype phase to comply with ISO guidelines, utilizing a hardware relay strictly for a scale-model cutoff demonstration.
- **Communication:** Dual-band strategy. LoRaWAN handles low-bandwidth, long-range macro-telemetry (fog trends, token requests). C-V2X or ESP-NOW handles high-rate, close-proximity Spatial Vector broadcasts.
- **Algorithms & AI:**
- **BrakeTwin Engine:** Derives maximum safe speed based on clear distance: $v_{safe} = a_{eff} \times [-t_{delay} + \sqrt{t_{delay}^2 + 2 \times D_{avail}/a_{eff}}]$.
- **Kinetic Priority Arbitration (KPA):** Replaces rigid "loaded truck wins" rules. Calculates hazard scores dynamically: $Score = (Mass \times Velocity) + (Mass \times g \times \sin(Gradient))$ to grant right-of-way to the vehicle with the most momentum.

4. Key Innovations

- **Spatial Vector Packet Protocol (SVPP):** Overcomes low bandwidth by having onboard AI convert video-detected hazards into 20-byte spatial vectors. Receiving trucks render these as "Phantom Obstacles" on their UI, effectively seeing through the truck ahead of them in zero visibility.
- **Safe-to-Proceed Corridor (D_clear ≥ D_req):** Replaces static distance alarms with a dynamic, physics-bound envelope that shrinks or grows based on real-time vehicle mass, downhill grade, and sensor health.
- **Idle-Truck Node Metamorphosis:** Maximizes sensing coverage without adding fixed infrastructure. When a truck idles at a crusher or loading bay, its software temporarily shifts into a predictive weather node, verifying past predictions and generating new ones.
- **RadarGlyph Anchoring:** Utilizes specific geometric arrangements of passive metal reflectors to create distinct radar signatures. This allows vehicles to identify critical zones and correct GNSS drift purely via radar returns.
- **Decentralized Emergency Stop Beacon:** A dedicated, single-button RF beacon carried by blast supervisors that broadcasts a highest-priority network-wide stop command, bypassing the need for a vulnerable centralized control room.

5. Feasibility

- **Prototype Feasibility:** Highly feasible for a 36-hour hackathon. The architecture deliberately relies on accessible ESP32/Pi hardware, 60GHz EVM radar boards, and RC carts to simulate heavy machinery.
- **Complexity:** Medium-High. The primary software complexity lies in the temporal synchronization of peer packets and the deterministic state machine.
- **Cost Drivers:** The 77GHz industrial radar units and RTK-GNSS modules dominate production costs. This is heavily offset by the "Idle-Truck Node" software feature, which drastically reduces the need for expensive fixed roadside infrastructure.
- **Real-World Deployment:** Scalable and realistic because it does not claim to execute autonomous braking on 100-ton dumpers. It operates strictly as an advisory system until integrated formally with OEM retarder/brake CAN interfaces under ISO 21815 and ISO 19014 standards.

6. Decision Matrix

Source / Feature	Status	Reason for Decision
Src 6: Safe-to-Proceed Corridor & BrakeTwin	Adopted	

Physics-based boundaries are mathematically auditable and far safer than opaque AI confidence scores or fixed-distance alarms.

Src 4: SVPP / Phantom Vision	Adopted	
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Solves the problem of transmitting hazard awareness over low-bandwidth mine RF networks.

Src 4: Multi-Modal Fog/Dust Discriminator	Adopted	
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Solves the critical real-world problem of mine dust triggering false fog alarms.

Src 2: Idle-Truck Nodes & Self-Verification	Adopted	
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Dramatically expands network coverage and self-corrects sensor drift at zero hardware cost.

Src 6: RadarGlyph Passive Anchors	Adopted	
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Solves GNSS multipath/canyon failures common in deep open-pit mines.

Src 1/6: Virtual Block / Movement Tokens	Adopted	
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Safely arbitrates single-lane blind curves with a fail-closed methodology.

Src 4: KPA (Kinetic Priority Arbitration)	Adopted	
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Provides a mathematically superior right-of-way resolution compared to human guesswork or static rules.

Src 3: Active Worker Tags	Adopted	
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Essential for pedestrian safety in low visibility where optical/radar cross-sections are minimal.

Src 2: Decentralized Emergency Beacon	Adopted	
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Superior to a centralized control room for immediate, localized blast-zone halting.

Src 1: Alpha Escort Mode	Rejected	
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Having a truck with failed sensors blindly follow another truck introduces unacceptable tailgating risks. System must fail to a safe stop instead.

Src 4: Ultrasonic as primary	Rejected	
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Insufficient range for heavy machinery. Upgraded to 77GHz FMCW Radar as the primary ranging tool.

7. Critical Weaknesses & Mitigation

- **Risk:** *GNSS Multipath in Deep Pits.* Walls reflect satellite signals, causing the truck to appear on the wrong map segment, corrupting gradient assumptions.
- **Mitigation:** The system relies on IMU dead-reckoning during outages and uses passive RadarGlyphs for absolute position anchoring without satellites.
- **Risk:** *RF Network Interference/Loss.* Heavy terrain blocks V2V signals, causing blind spots.
- **Mitigation:** The core safety loop is offline-first. If the network drops, the vehicle strictly defaults to its onboard radar, shortens its trusted corridor, and caps its speed.
- **Risk:** *Driver Alarm Fatigue.* Constant pinging from dust, minor hazards, and distant trucks causes drivers to ignore the system.
- **Mitigation:** The Trust Engine aggregates evidence. The UI presents only one authoritative alert state and one recommended speed, suppressing duplicate alarms.

8. Implementation Roadmap

Phase	Milestone	Deliverables & Testing Focus
1. Prototype (Hackathon)	36-Hour Core Logic Validation	

Two RC carts, one ESP32 fog node, SVPP mesh communication. Prove the BrakeTwin math locally and successfully fail-closed during network loss.

2. Closed-Track Pilot	Sensor Range & Vibration	
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Mount industrial 77GHz radar and edge compute on a light vehicle. Validate radar penetration in artificial fog and RadarGlyph detection against mine clutter.

3. Shadow Fleet Pilot	Passive Advisory Logging	
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Install on 5-10 dumpers. System runs silently, logging D_clear vs D_req against actual driver behavior to tune the Trust Engine and minimize nuisance alerts.

4. Advisory Deployment	Active HMI Guidance	
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Activate driver dashboard. Install fixed Multi-Modal nodes at known fog hotspots. Establish Virtual Blocks at critical blind curves.

5. OEM Integration	ISO Compliant Actuation	
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Partner with machine OEMs to map the SafeCorridor engine to the vehicle's retarder/brake CAN bus for automatic intervention as a last resort.